

SUB QUERIES

--NESTED QUERY OR QUERY WITH IN A QUERY--

---TYPES OF SUBQUERY---

--1 SINGLE ROW SUBQUERY (SCALAR VALUE)

--2 MUTIROW SUBQUERY

--3 CORRELATED SUBQUERY

```
CREATE TABLE EMP(  
    EMPLOYEE_ID INT PRIMARY KEY,  
    EMPLOYEE_NAME VARCHAR(100),  
    SALARY DECIMAL(10, 2),  
    MANAGER_ID INT,  
    DEPARTMENT_ID INT,  
    HIRE_DATE DATE,  
    CITY VARCHAR(100)  
);  
INSERT INTO EMP(EMPLOYEE_ID, EMPLOYEE_NAME, SALARY, MANAGER_ID, DEPARTMENT_ID, HIRE_DATE, CITY)  
VALUES  
(1, 'RAJESH KUMAR', 75000.00, NULL, 1, '2010-01-15', 'MUMBAI'),  
(2, 'PRIYA SHARMA', 85000.00, 1, 2, '2011-03-20', 'DELHI'),  
(3, 'ANITA SINGH', 95000.00, 1, 1, '2012-05-30', 'BANGALORE'),  
(4, 'VIKAS PATEL', 70000.00, 3, 3, '2013-07-10', 'CHENNAI'),  
(5, 'SUNIL VERMA', 60000.00, 3, 2, '2014-09-25', 'HYDERABAD'),  
(6, 'NEHA GUPTA', 90000.00, 2, 1, '2015-11-05', 'PUNE'),  
(7, 'AMIT JAIN', 65000.00, 4, 3, '2016-01-20', 'KOLKATA'),  
(8, 'KAVITA RAO', 80000.00, 2, 1, '2017-03-15', 'MUMBAI'),  
(9, 'RAHUL DESAI', 72000.00, 1, 3, '2018-05-10', 'DELHI'),  
(10, 'SUMAN MISHRA', 82000.00, 4, 2, '2019-07-30', 'BANGALORE'),  
(11, 'ANITA SINGH', 95000.00, 1, 1, '2012-05-30', 'BANGALORE');
```

```
select * from emp
```

--SINGLE ROW SUBQUERY

--1 RETRIEVE THE NAME OF THE EMPLOYEE WITH THE HIGHEST SALARY

--ERROR

```
SELECT EMPLOYEE_NAME, MAX(SALARY) FROM EMP
```

```
SELECT MAX(SALARY) FROM EMP
```

--95000

```
SELECT EMPLOYEE_NAME, SALARY
```

```
FROM EMP
```

```
WHERE SALARY =95000
```

```
SELECT EMPLOYEE_NAME, SALARY
```

```
FROM EMP
```

```
WHERE SALARY = (SELECT MAX(SALARY) FROM EMP);
```

--2 GET THE NAME OF THE EMPLOYEE WITH THE LOWEST SALARY

```
SELECT EMPLOYEE_NAME ,SALARY
```

```
FROM EMP
```

```
WHERE SALARY = (SELECT MIN(SALARY) FROM EMP);
```

--3 FIND THE EMPLOYEE WITH THE EARLIEST HIRE DATE

```
--RETRIEVE THE NAME OF THE EMPLOYEE WHO HAS BEEN  
--WORKING THE LONGEST (BY HIRE_DATE)
```

```
--MIN --> OLDEST DATE  
--MAX--> LATEST OR MOST RECENT DATE
```

```
SELECT EMPLOYEE_NAME, HIRE_DATE  
FROM EMP  
WHERE HIRE_DATE = (SELECT MIN(HIRE_DATE) FROM EMP);
```

```
--4 FIND THE EMPLOYEE WHO IS RECENTLY HIRED  
SELECT EMPLOYEE_NAME, HIRE_DATE  
FROM EMP  
WHERE HIRE_DATE = (SELECT MAX(HIRE_DATE) FROM EMP);
```

```
--5 GET EMPLOYEES WHO EARN MORE THAN THE AVERAGE SALARY;
```

```
SELECT EMPLOYEE_NAME, SALARY  
FROM EMP  
WHERE SALARY > (SELECT AVG(SALARY) FROM EMP);
```

```
select * from emp  
--6 FIND EMPLOYEES IN THE SAME DEPARTMENT AS 'NEHA GUPTA'  
SELECT *  
FROM EMP  
WHERE DEPARTMENT_ID =  
(SELECT DEPARTMENT_ID FROM EMP WHERE EMPLOYEE_NAME='NEHA GUPTA')
```

```
-----FREQUENTLY ASKED INTERVIEW QUERY-----
```

```
--7 FIND SECOND HIGHEST SALARY  
select salary from emp  
---METHOD:1  
--STEP 1: ALL THE SALARY OTHER THEN MAX SALARY  
SELECT SALARY  
FROM EMP  
WHERE SALARY < (SELECT MAX(SALARY) FROM EMP)  
--95000
```

```
--STEP 2 FIND MAX SALARY FROM OTHER THEN MAX SALARY  
SELECT MAX(SALARY)  
FROM EMP  
WHERE SALARY < (SELECT MAX(SALARY) FROM EMP)
```

```
---METHOD:2  
--WITH ORDER BY  
SELECT SALARY  
FROM EMP  
WHERE SALARY < (SELECT MAX(SALARY) FROM EMP)  
ORDER BY SALARY DESC
```

```
---THEN SELECT TOP 1  
SELECT TOP 1 SALARY  
FROM EMP
```

```
WHERE SALARY < (SELECT MAX(SALARY) FROM EMP)
ORDER BY SALARY DESC
```

```
--8 SELECT N TH HIGHEST SALARY
--LET'S CONSIDER 4TH HIGHEST SALARY
```

```
--STEP 1
SELECT SALARY
FROM EMP
ORDER BY SALARY DESC
```

```
--STEP 2
SELECT DISTINCT TOP 4 SALARY
FROM EMP
ORDER BY SALARY DESC
```

```
--STEP 3
SELECT TOP 1 SALARY FROM
    (SELECT DISTINCT TOP 4 SALARY
     FROM EMP
     ORDER BY SALARY DESC) AS RESULT
ORDER BY SALARY
```

```
---OR---
SELECT MIN(SALARY) FROM
    (SELECT DISTINCT TOP 4 SALARY
     FROM EMP
     ORDER BY SALARY DESC) AS RESULT
```

```
SELECT EMPLOYEE_NAME, SALARY, CITY FROM EMP WHERE
SALARY =
(SELECT MAX(SALARY) FROM EMP)
```

```
--9 FIND THE EMPLOYEE WORKING IN THE SAME CITY
--AS THE HIGHEST-PAID EMPLOYEE
```

```
SELECT EMPLOYEE_NAME, CITY, SALARY
FROM EMP
WHERE CITY = (
    SELECT DISTINCT CITY
    FROM EMP
    WHERE SALARY = (SELECT MAX(SALARY) FROM EMP)
);
```

```
select * from emp
```

```
SELECT EMPLOYEE_NAME, CITY, SALARY
FROM EMP
WHERE CITY = (
    SELECT top 1 CITY
    FROM EMP
    order by salary desc)
```

```
-----
--MULTI ROW SUBQUERY
-----
```

```
--GET EMPLOYEES WHO WORK IN DEPARTMENTS WHERE THE AVERAGE SALARY
```

--OF DEPT IS ABOVE 70,000

```
SELECT DEPARTMENT_ID,AVG(SALARY)
FROM EMP
GROUP BY DEPARTMENT_ID
HAVING AVG(SALARY)>70000
```

```
SELECT EMPLOYEE_NAME,SALARY,DEPARTMENT_ID
FROM EMP
WHERE DEPARTMENT_ID IN
    (SELECT DEPARTMENT_ID
    FROM EMP
    GROUP BY DEPARTMENT_ID
    HAVING AVG(SALARY)>70000)

--1 2
```

--GET EMPLOYEE NAME WHO WHORKS IN A DEPARTMENT WHERE
--TOTAL EMPLOYEE IN DEPT IS MORE THAN 2

```
--STEP 1
SELECT DEPARTMENT_ID,count(*)
FROM EMP
GROUP BY DEPARTMENT_ID
HAVING COUNT(EMPLOYEE_NAME)>2
```

```
--STEP 2
SELECT EMPLOYEE_NAME
FROM EMP
WHERE DEPARTMENT_ID IN (SELECT DEPARTMENT_ID
    FROM EMP
    GROUP BY DEPARTMENT_ID
    HAVING COUNT(EMPLOYEE_NAME)>2)

--1 2 3
```

--CORELATED ROW SUBQUERY

```
CREATE TABLE EMPLOYEEMASTER
(EID INT,
NAME VARCHAR(10))
```

```
INSERT INTO EMPLOYEEMASTER VALUES
(1,'JAY'),(2,'MEET'),(3,'MINAL'),(4,'MIRA'),(5,'RAHUL')
```

```
CREATE TABLE DEPTMASTER
(DEPT_ID VARCHAR(5),
DEPT_NAME VARCHAR(10),
EID INT)
```

```
INSERT INTO DEPTMASTER VALUES
('D1','IT',1),('D2','HR',2),('D3','MARKETING',3)
```

```
SELECT * FROM EMPLOYEEMASTER
SELECT * FROM DEPTMASTER
```

--THE EXISTS OPERATOR IS USED TO TEST FOR THE EXISTENCE OF ANY RECORD IN A SUBQUERY.

--THE EXISTS OPERATOR RETURNS TRUE IF THE SUBQUERY RETURNS ONE OR MORE RECORDS.

```
SELECT * FROM EMPLOYEEMASTER
WHERE EXISTS
    (SELECT EID FROM DEPTMASTER
     WHERE
        EMPLOYEEMASTER.EID=DEPTMASTER.EID)
```